Assignment-3

(Character Functions)

1. Write a query to display all the employees name , Designation & Email in Upper case who are getting salary more than 12000 .

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```
SELECT UPPER(first_name || ' ' || last_name) AS employee_name,
       UPPER(job_id) AS designation,
       UPPER(email) AS email
FROM employees
WHERE salary > 12000
```

Results Explain Describe Saved SQL History

EMPLOYEE_NAME	DESIGNATION	EMAIL
STEVEN KING	AD_PRES	SKING
NEENA KOCHHAR	AD_VP	NKOCHHAR
LEX DE HAAN	AD_VP	LDEHAAN
JOHN RUSSELL	SA_MAN	JRUSSEL
KAREN PARTNERS	SA_MAN	KPARTNER
MICHAEL HARTSTEIN	MK_MAN	MHARTSTE

6 rows returned in 0.00 seconds CSV Export

2. Write a query to display all the employees name , Designation & Email in lower case who are getting salary less than 12000.

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```
SELECT LOWER(first_name || ' ' || last_name) AS employee_name,
       LOWER(job_id) AS designation,
       LOWER(email) AS email
FROM employees
WHERE salary < 12000
```

Results Explain Describe Saved SQL History

EMPLOYEE_NAME	DESIGNATION	EMAIL
alexander hunold	it_prog	ahunold
bruce ernst	it_prog	bernst
david austin	it_prog	daustin
valli pataballa	it_prog	vpatabal
diana lorentz	it_prog	dlorentz
daniel faviet	fi_account	dfaviet
john chen	fi_account	jchen
ismael sciarra	fi_account	isciarra
jose manuel urman	fi_account	jmurman
luis popp	fi_account	lpopp
den raphaely	pu_man	drapheal
kevin feeney	sh_clerk	kfeeney
donald oconnell	sh_clerk	doconnel
douglas grant	sh_clerk	dgrant
jennifer whalen	ad_asst	jwhalen
pat fay	mk_rep	pfay
susan mavris	hr_rep	smavris
hermann baer	pr_rep	hbaer
william gietz	ac_account	wgietz

98 rows returned in 0.02 seconds CSV Export

3. Write a query to display Name, designation, salary of employees such that salary should be padded with '*' from left for department 60, 90,100.

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```
SELECT first_name || ' ' || last_name AS employee_name,
       job_id AS designation,
       LPAD(salary, 10, '*') AS padded_salary
FROM employees
WHERE department_id IN (60, 90, 100)
```

Results Explain Describe Saved SQL History

EMPLOYEE_NAME	DESIGNATION	PADDED_SALARY
Steven King	AD_PRES	*****24000
Neena Kochhar	AD_VP	*****17000
Lex De Haan	AD_VP	*****17000
Alexander Hunold	IT_PROG	******9000
Bruce Ernst	IT_PROG	******6000
David Austin	IT_PROG	******4800
Valli Pataballa	IT_PROG	******4800
Diana Lorentz	IT_PROG	******4200
Nancy Greenberg	FI_MGR	*****12000
Daniel Faviet	FI_ACCOUNT	******9000
John Chen	FI_ACCOUNT	******8200
Ismael Sciarra	FI_ACCOUNT	******7700
Jose Manuel Urman	FI_ACCOUNT	******7800
Luis Popp	FI_ACCOUNT	******6900

14 rows returned in 0.02 seconds CSV Export

4. Write a query to display Name, designation, salary of employees such that salary should be padded with '#' from right for department 80.

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```
SELECT first_name || ' ' || last_name AS employee_name,
       job_id AS designation,
       RPAD(salary, 10, '#') AS padded_salary
FROM employees
WHERE department_id = 80
```

Results Explain Describe Saved SQL History

EMPLOYEE_NAME	DESIGNATION	PADDED_SALARY
John Russell	SA_MAN	14000######
Karen Partners	SA_MAN	13500######
Alberto Errazuriz	SA_MAN	12000######
Gerald Cambrault	SA_MAN	11000######
Eleni Zlotkey	SA_MAN	10500######
Peter Tucker	SA_REP	10000######
David Bernstein	SA_REP	9500#######
Peter Hall	SA_REP	9000#######
Christopher Olsen	SA_REP	8300#######
Elizabeth Bates	SA_REP	7300#######
Sundita Kumar	SA_REP	6100#######
Ellen Abel	SA_REP	11000######
Alyssa Hutton	SA_REP	8800#######
Jonathon Taylor	SA_REP	8600#######
Jack Livingston	SA_REP	8400#######
Charles Johnson	SA_REP	6200#######

34 rows returned in 0.01 seconds CSV Export

5. Write a query to display The name , salary, commission and total salary of each employee who are getting salary more than 10000 . Name should be like S. KING, D.AUSTIN, etc.

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```
SELECT SUBSTR(first_name, 1, 1) || '. ' || UPPER(last_name) AS employee_name,
       salary,
       commission_pct,
       salary + (salary * NVL(commission_pct, 0)) AS total_salary
FROM employees
WHERE salary > 10000
```

Results Explain Describe Saved SQL History

EMPLOYEE_NAME	SALARY	COMMISSION_PCT	TOTAL_SALARY
S. KING	24000	-	24000
N. KOCHHAR	17000	-	17000
L. DE HAAN	17000	-	17000
N. GREENBERG	12000	-	12000
D. RAPHAELY	11000	-	11000
J. RUSSELL	14000	.4	19600
K. PARTNERS	13500	.3	17550
A. ERRAZURIZ	12000	.3	15600
G. CAMBRAULT	11000	.3	14300
E. ZLOTKEY	10500	.2	12600
C. VISHNEY	10500	.25	13125
L. OZER	11500	.25	14375
E. ABEL	11000	.3	14300
M. HARTSTEIN	13000	-	13000
S. HIGGINS	12000	-	12000

15 rows returned in 0.01 seconds CSV Export

Note: ***NVL*** is used to convert null commissions to 0 before calculating total salary.

6. Write a query to display employees name by removing the 1st character from both First_name & Last_name.

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```
SELECT SUBSTR(first_name, 2) || ' ' || SUBSTR(last_name, 2) AS trimmed_name
FROM employees
```

Results Explain Describe Saved SQL History

TRIMMED_NAME
teven ing
eena ochhar
ex e Haan
lexander unold
ruce rnst
avid ustin
alli ataballa
ichael artstein
at ay
usan avris
ermann aer
helley iggins
illiam ietz

107 rows returned in 0.01 seconds CSV Export

SUBSTR Function in SQL

Syntax:

SUBSTR(string, start_position, length**)**

Meaning:

- **string** → The column or text you're working with.
- **start_position** → The position in the string where extraction begins.
  - Positions start at **1** (not 0).
- **length** *(optional)* → How many characters to extract.
  - If omitted, it takes everything from start_position to the end of the string.

Examples:

- **SUBSTR(last_name, 2)**
  → Starts from the second character of last_name.
  Example: Smith → mith

- **SUBSTR(first_name, 1, 1)**
  → Extracts the first character of first_name.
  Example: John → J

7. Write a query to display the Name and their short form of All employees such as Steven King = S.K.

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```
SELECT first_name || ' ' || last_name AS full_name,
       SUBSTR(first_name, 1, 1) || '.' || SUBSTR(last_name, 1, 1) || '.' AS short_form
FROM employees
```

Results Explain Describe Saved SQL History

FULL_NAME	SHORT_FORM
Steven King	S.K.
Neena Kochhar	N.K.
Lex De Haan	L.D.
Alexander Hunold	A.H.
Bruce Ernst	B.E.
David Austin	D.A.
Valli Pataballa	V.P.
Michael Hartstein	M.H.
Pat Fay	P.F.
Susan Mavris	S.M.
Hermann Baer	H.B.
Shelley Higgins	S.H.
William Gietz	W.G.

107 rows returned in 0.02 seconds CSV Export

8. Write a query to display the Location ID , city, state_ province and city code where city code is the first four character of city, in descending order of city code for country id US,UK,CA,JP and IN.

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```
SELECT location_id,
       city,
       state_province,
       SUBSTR(city, 1, 4) AS city_code
FROM locations
WHERE country_id IN ('US', 'UK', 'CA', 'JP', 'IN')
ORDER BY city_code DESC
```

Results Explain Describe Saved SQL History

LOCATION_ID	CITY	STATE_PROVINCE	CITY_CODE
1900	Whitehorse	Yukon	Whit
1800	Toronto	Ontario	Toro
1200	Tokyo	Tokyo Prefecture	Toky
2600	Stretford	Manchester	Stre
1400	Southlake	Texas	Sout
1500	South San Francisco	California	Sout
1600	South Brunswick	New Jersey	Sout
1700	Seattle	Washington	Seat
2500	Oxford	Oxford	Oxfo
2400	London	-	Lond
1300	Hiroshima	-	Hiro
2100	Bombay	Maharashtra	Bomb

12 rows returned in 0.00 seconds CSV Export

9. Write a query to display the Department ID, Department Name, department code , Location ID which having manager. Department code is the First 2 Characters from department name.

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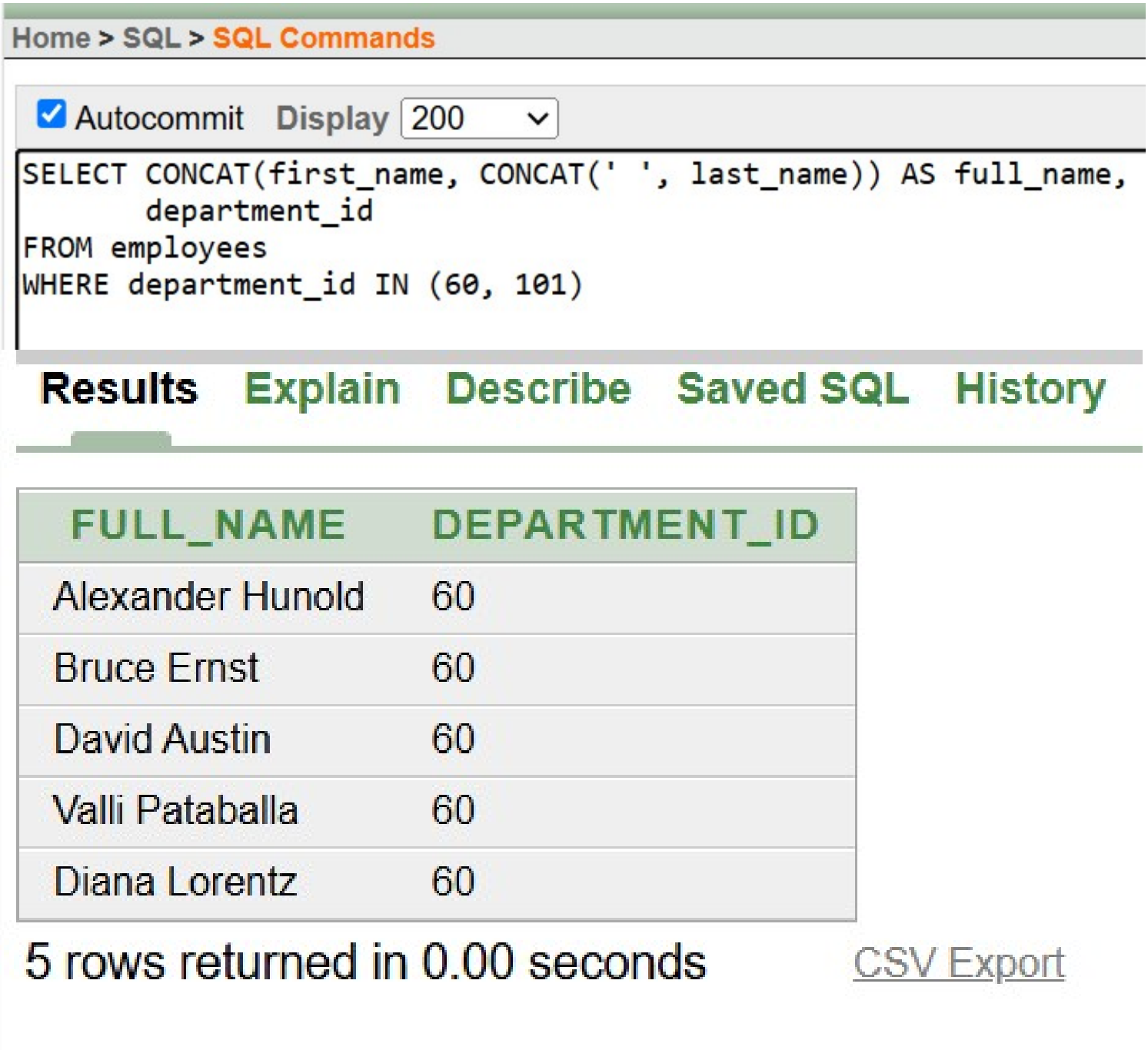
```
SELECT department_id,
       department_name,
       SUBSTR(department_name, 1, 2) AS department_code,
       location_id
FROM departments
WHERE manager_id IS NOT NULL
```

Results Explain Describe Saved SQL History

DEPARTMENT_ID	DEPARTMENT_NAME	DEPARTMENT_CODE	LOCATION_ID
10	Administration	Ad	1700
20	Marketing	Ma	1800
30	Purchasing	Pu	1700
40	Human Resources	Hu	2400
50	Shipping	Sh	1500
60	IT	IT	1400
70	Public Relations	Pu	2700
80	Sales	Sa	2500
90	Executive	Ex	1700
100	Finance	Fi	1700
110	Accounting	Ac	1700

11 rows returned in 0.02 seconds CSV Export

10. Write a query to display the fullname and department_id of employees who are working in department 101 and 60 . joining first_name and last_name using function. There should be a space between First and last name.



11. Write a query to display the name, job_id and salary of each employee padded with '#' from both left and right who contain 'IT_' in their job_ID.

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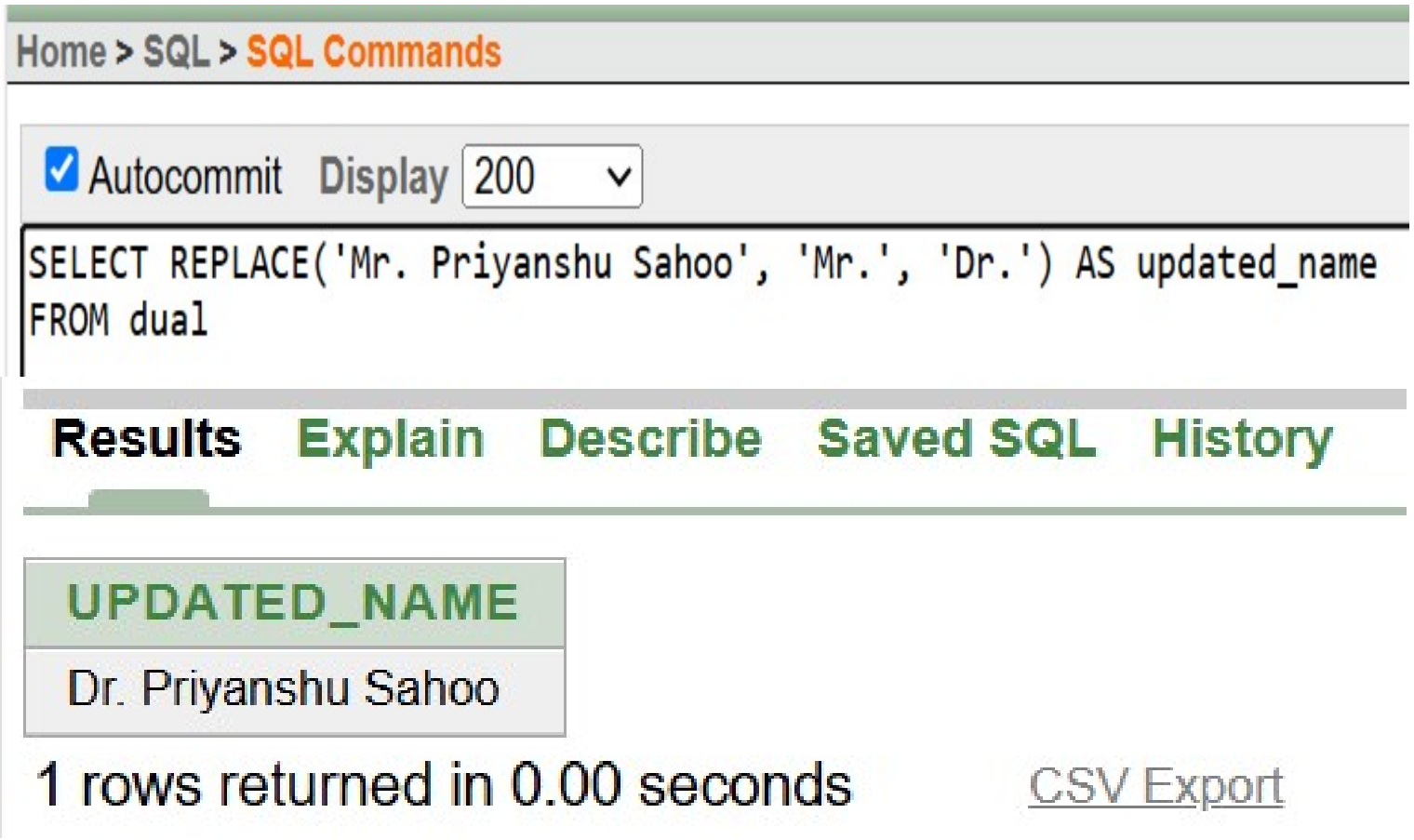
```
SELECT first_name || ' ' || last_name AS employee_name,
       job_id,
       RPAD(LPAD(salary, LENGTH(salary) + 3, '#'), LENGTH(salary) + 6, '#') AS padded_salary
FROM employees
WHERE job_id LIKE '%IT_%'
```

Results Explain Describe Saved SQL History

EMPLOYEE_NAME	JOB_ID	PADDED_SALARY
Alexander Hunold	IT_PROG	###9000###
Bruce Ernst	IT_PROG	###6000###
David Austin	IT_PROG	###4800###
Valli Pataballa	IT_PROG	###4800###
Diana Lorentz	IT_PROG	###4200###

5 rows returned in 0.00 seconds CSV Export

12. Write a query to replace the title of your name with another title.



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